Airbnb Lense

Capturing Your Style

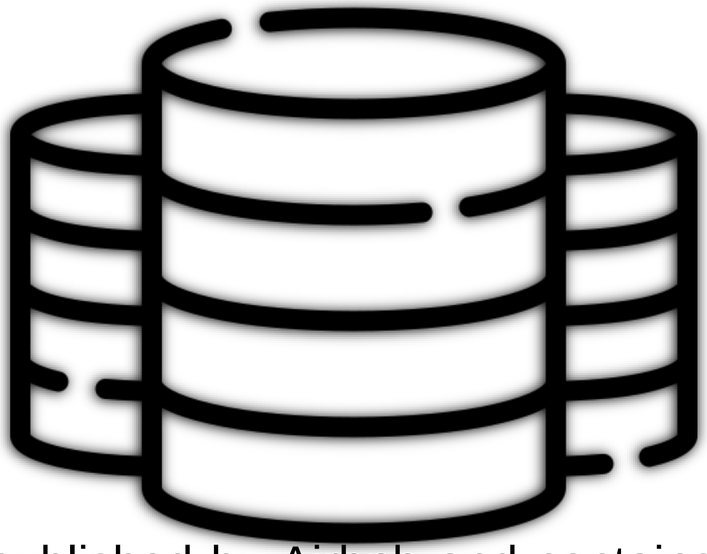
Harsh Shroff | harshroff@umbc.edu
DATA 606 - Dr Chaojie (Jay) Wang

Introduction

The project is about the development of an Airbnb Photo & Text-Based Recommendation System. This system aims to provide Airbnb users with personalized recommendations for accommodations based on the photos of their previous stays and/or by inputing a couple of sentences. It utilizes computer vision and machine learning techniques to analyze images and match them with similar Airbnb listings. It also uses an OpenAi API to understand the text input and get the recommendations. The system benefits both travelers looking for unique and stylish accommodations and hosts seeking insights into their competition.

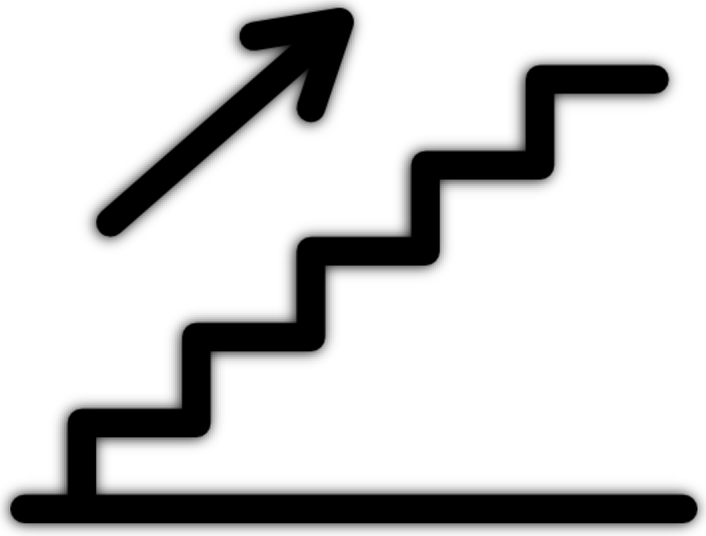
Data Description

- CSV Files: 160 MB
- Images per state:
  - New York (NYC): 80.5 GB
  - North Carolina (Asheville): 9.44 GB
  - Rhode Island (Rhode Island): 11.5 GB
  - District of Columbia (Washington): 14.3 GB
  - New Jersey (Newark + Jersey City): 5.36 GB
  - Massachusetts (Boston + Cambridge): 9.48 GB
- The data source for this project is the InsideAirbnb dataset, which is published by Airbnb and contains comprehensive information about Airbnb listings worldwide, including property details, host information, and photos.



Project Methodology

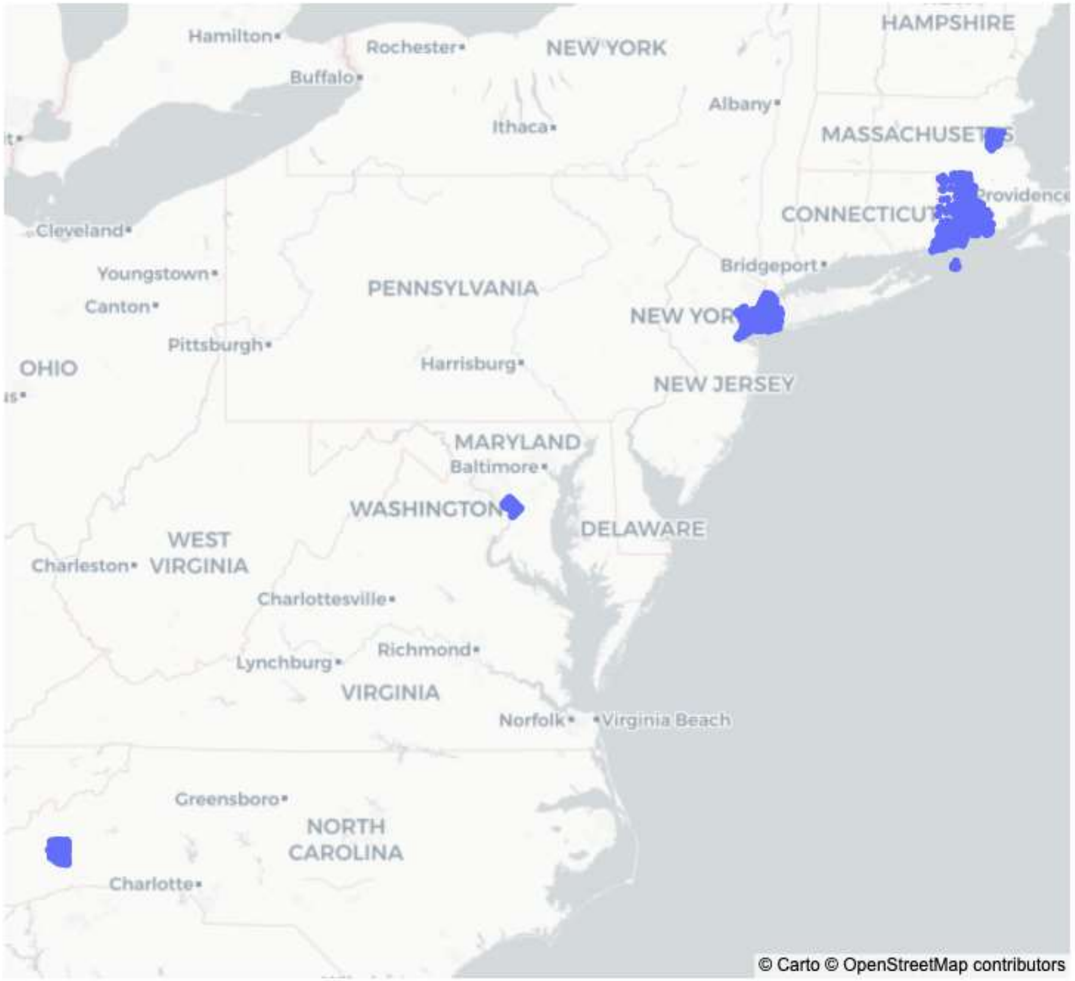
1. Data Preprocessing & EDA
2. ML Model Training & Testing
3. Model Deployment on a Streamlit Webapp



Data Preprocessing & EDA

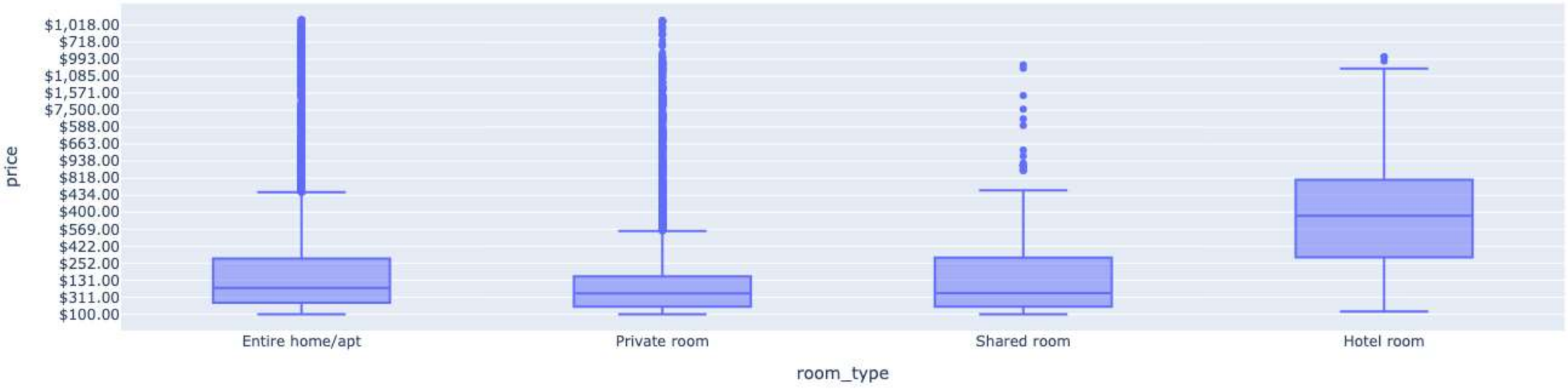
Map of Listings

- The map generated using the latitude and longitude data available in the .csv file shows listings in all cities and states.



Price Distribution

Price Distribution by Room Type



ML Model

Model Summary

- As seen, the model is not trainable and is used for transfer learning.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 resnet152v2 (Functional)    (None, 7, 7, 2048)        58331648

 global_max_pooling2d (Globa  (None, 2048)             0
 lMaxPooling2D)

=================================================================
Total params: 58,331,648
Trainable params: 0
Non-trainable params: 58,331,648
_________________________________________________________________
None
```

Deployment

Text Based Recommender

- Gives three recommendations with respect to given text.

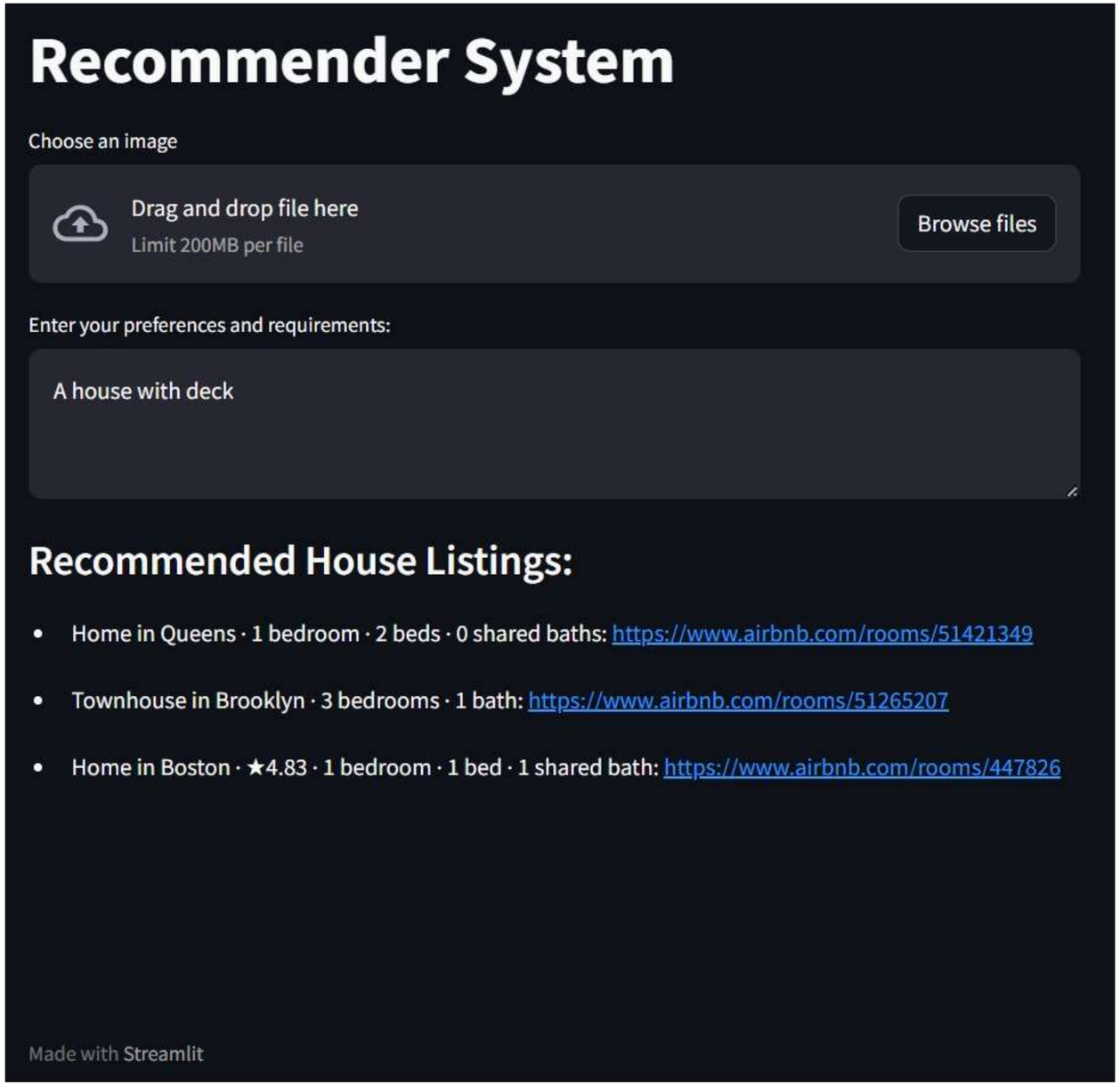
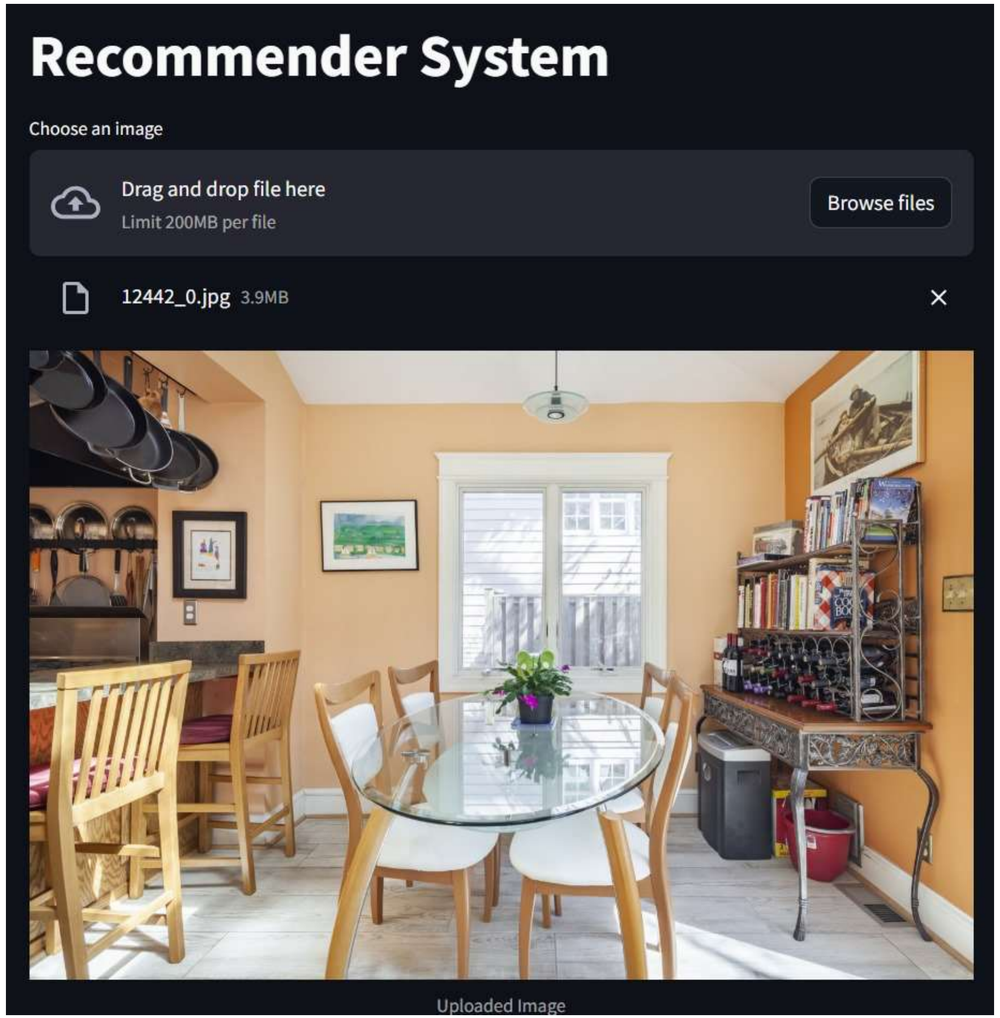
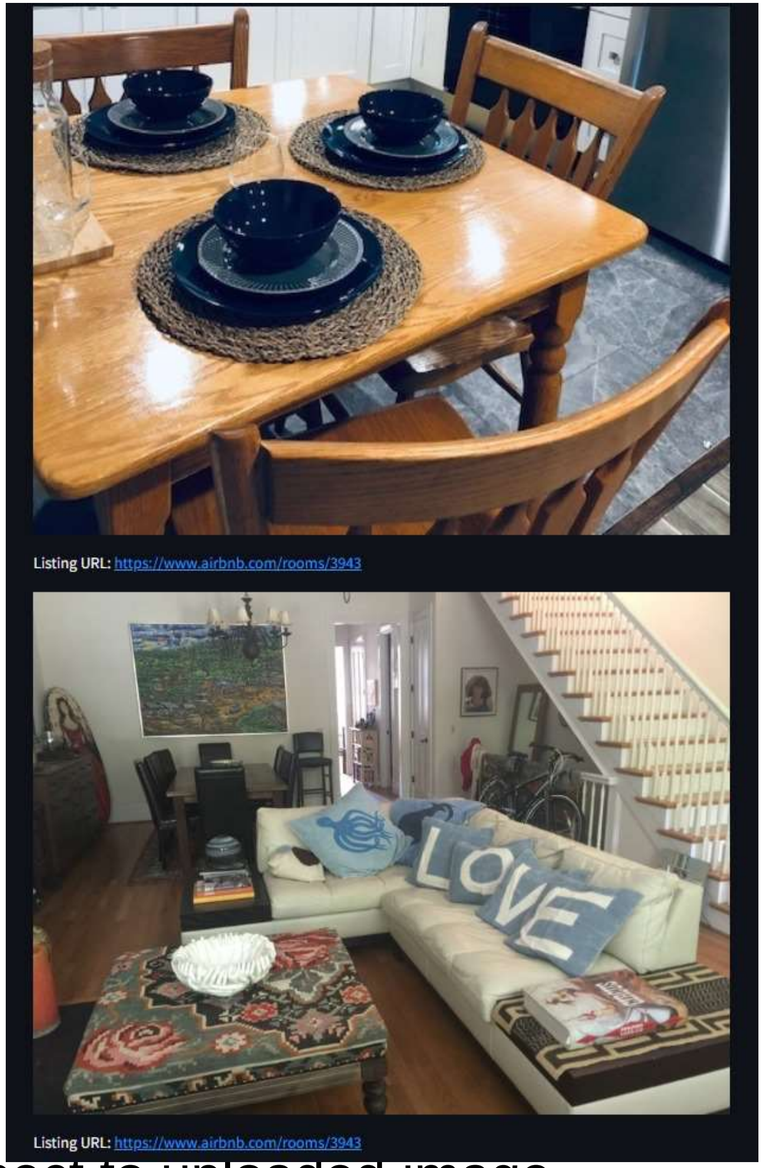


Image Based Recommender





Gives three recommendations with respect to uploaded image.

Thank You!

Harsh Shroff | harshroff@umbc.edu
DATA 606 - Dr Chaojie (Jay) Wang